Frequency and spectrum of mutations in the BRCA1, BRCA2, PALB2, P53, PTEN, CHEK2, CDH1 genes in women with breast cancer from three cities of Colombia

Alejandro Mejía García[1]; Luz María Gonzales[1]; William Hernán Arias Pérez MSc[2]; Yina Tatiana Zambrano[1]; Shalom Gómez Pugarin[1]; Johanna Tejada Moreno[2]; Roberto Jaramillo[3], Yorlany Rodas[3], Edgar Navarro[4], Andrés Ossa[5], Mauricio Borrero, Gonzalo Alberto Angel, MSc[2] Alicia M. Cock-Rada[6] MSc. Gabriel Bedoya[2] MSc; Michael Dean[7], Sabina Rinaldi Msc, PhD[8] Gloria Inés Sánchez MSc. PhD.[1]

[1]Grupo Infección y Cáncer. [2]Grupo Genética Molecular. Universidad de Antioquia, Medellín, Colombia, [3] Hemato oncólogos S.A. [4]Universidad del Norte, Barranquilla, Colombia. [5] hospital general de Medellín, [6] instituto de Cancerología Las Américas, Medellín, Colombia. [7]National institue of health,, Bethesda, United states, [8]International agency for research on cancer

Introduction

5-10% of breast cancers are hereditary, genetic studies have found causal mutations in genes such as BRCA1, BRCA2, PALB2, etc. In Colombia there are diverse mutations, the most frequent are the founder variants A1708E(BRCA1), 3450delCAAG y 3034delACAA(BRCA2)(1).

The frequency of pathogenic variants have been evaluated in unselected women only in a few studies, with results of 1.2% - 4.2 % (2, 3). However, the frequency depends on several factors such as the presence of founder mutations, the number of genes evaluated, the technique for the detection and the selection criteria of cases.

Author	Population	Frequency
Malone KE et al (4) J Peto et al (4)	USA UK	BRCA1: 0,5-3,1% BRCA2: 0,8-3%
Malone KE (4)	African-american	BRCA1: 1,3-1,4% BRCA2: 2,6%
EM John et al (4)	Hispanic	BRCA1: 3,5%
Hernandez et al (5)	Medellín	1,2% (BRCA)
Sun J et al (6)	China	9.2% (62 genes)

AIM

To determine the frequency and spectrum of mutations in the BRCA1, BRCA2, PALB2, P53, PTEN, CHEK2, CDH1 genes in women with breast cancer from three cities of Colombia (Medellín, Cali and Barranquilla)

Methods

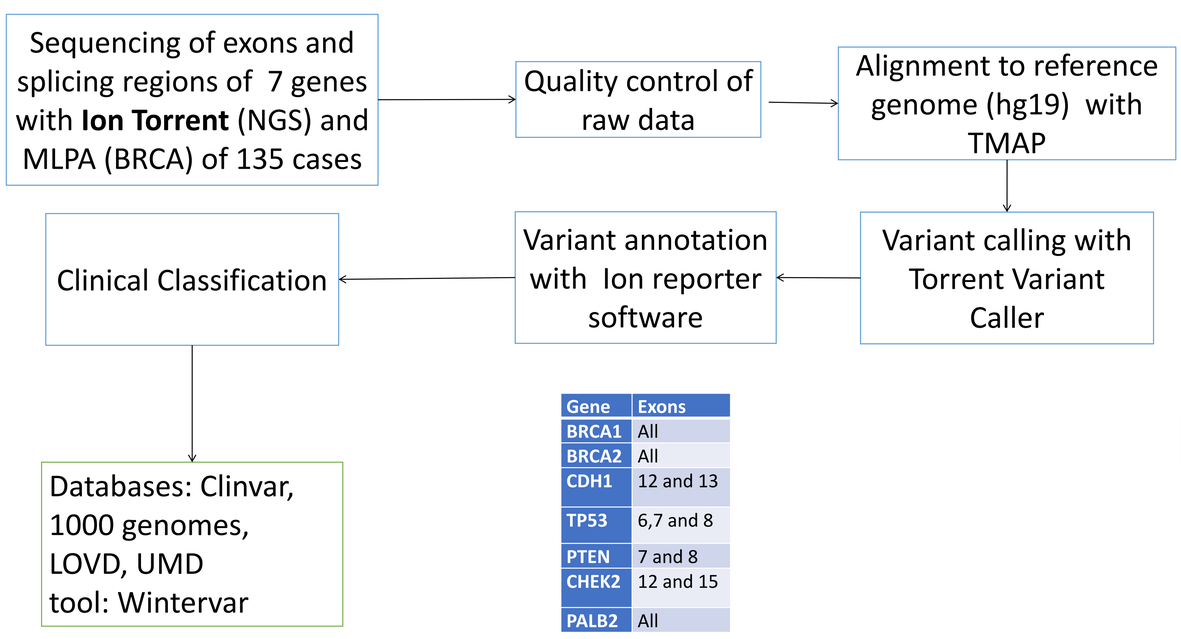
Population

Women with breast cancer that assisted to one of the 6 health centers of the study in Medellín, Cali and Barranquilla between 2012-2016

Inclusion criteria

- incident cases of primary breast cancer (first diagnosis), including invasive and in situ breast cancers. She should be recruited at referral for diagnostic biopsy.
- She will have resided in the same area as the study for at least 3 years.
- Criterios de exclusión:
- She will have a positive biopsy for breast cancer with clinical staging according to TNM standards.

Mutation Detection



Gene	Exons
BRCA1	All
BRCA2	All
CDH1	12 and 13
TP53	6,7 and 8
PTEN	7 and 8
CHEK2	12 and 15
PALB2	All

Results

Frequency of pathogenic mutations carriers = 4.4% (6/135)

Table 1. Description of pathogenic variants found in the patients

Gene	ID	DNA change	Aminoacide change	Type of variant	Clinvar	LOVD	Wintervar	UMD	Reported in Colombia
TP53	rs397516435	c.586C>T	p.Arg196Ter	Nonsense	P	-	P	-	No
BRCA1	rs28897696	c.5123C>A	p.Ala1708Glu	Missense	P	-	LP	P	Yes
BRCA1	rs80357471	c.178C>T	p.Gln60Ter	Nonsense	P	-	P	P	No
BRCA2	rs397507891	c.7007+1G>A	NA	intronic	P	-	-	VUS	No
BRCA1	rs80358163	c.213-12A>G	NA	Intronic	P	P	-	-	Yes
BRCA2	rs397507840	c.631+3A>G	NA	intronic	P (1), VUS (2)	P	-	P	No

P= pathogenic

LP= Likely pathogenic

VUS= Variant of uncertain significance

Table 2. classification of VUS and effect with in silico predictors

dbSNP	HGVS	tipo	ClinVar	wintervar	LOVD	FATHMM	POLYPHEN2_HVAR	PROVEAN	REVEL
BRCA1									
NA	c.4097-21C>T	intronic	-	-	1(2), 3(1)	NA	NA	NA	
rs45489593	c.441+41C>T	intronic	3 (1)	-	2(1)	NA	NA	NA	
rs397507199	c.2329T>G	Missense	1(1), 2(2), 3(2)	3	NA	T	B	**D**	**0.514**
BRCA2									
rs749729890	c.773A>G	Missense	3 (1)	3 (1)	NA	T	B	N	0.197
NA	c.855G>T	Missense	-	3 (1)	NA	T	P	N	0.117
NA	c.2129C>T	Missense	-	3	NA	T	B	**D**	0.219
rs80358575	c.3299A>T	Missense	2,3 (9)	2	3	T	P	**D**	0.079
rs397507306	c.3321A>C	Missense	3 (4)	3	NA	T	**D**	**D**	0.180
NA	c.5803A>T	Missense	-	3	NA	T	B	**D**	0.157
rs80358880	c.6443C>A	Missense	1,2,3 (9)	2	NA	T	B	N	**0.532**
rs56204128	c.9353T>C	Missense	2(1), 3(9)	2	NA	T	B	N	**0.468**
PALB2									
rs587782151	c.1748T>G	Missense	3 (5)	2 (2)	3	T	**D**	**D**	0.113
CHEK2									
rs587782413	c.497A>G	Missense	3 (3)	3 (1)	NA	**D**	**D**	**D**	**0.821**
rs121908701	c.671G>A	Missense	3	3	1(2), 3(1)	**D**	B	N	**0.460**
CDH1									
rs34582129	c.1712-52G>C	intronic	-	-	3	NA	NA	NA	
rs35831514	c.1712-37T>G	intronic	-	-	3	NA	NA	NA	
rs780759537	c.1793G>A	Missense	3 (3)	3	3	T	B	N	0.125

NA: not available; T: tolerant, B: benign; D: deleterious. N: neutral. 1: benign; 2: likely benign; 3: variant of uncertain significance

Conclusions

- Frequency of carriers 4.4%, first study in Colombia that evaluated other genes than BRCA in unselected women
- Three splicing variants, so it is necessary to include intron-exon boundaries in the target
- The C.497A>G (CHEK2) variant has a great pathogenicity in silico, so it can be reclassified with in vitro studies.

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